改变人类命运的霉菌

The Mould That Changed the World



从一团霉菌到救命药，中间还差什么？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In September 1928 Alexander Fleming, a bacteriologist at St Mary's Hospital, London, returned from holiday to find that a stray mould had contaminated a dish of staphylococcus bacteria. Nothing grew around it. Popular mythology blames an open window; the spores probably originated in a laboratory downstairs, where a colleague cultivated moulds.

Fleming called the mould's juice penicillin and published in June 1929. Yet the crude juice was unstable and stubbornly hard to purify, and he and the chemists he enlisted lacked the expertise to refine it. He used it chiefly to isolate other microbes, and penicillin lay dormant, a curiosity rather than a cure.

A decade later, in Oxford, Howard Florey, an Australian pathologist, revived the work with Ernst Chain, a German-born chemist and Jewish refugee. Their colleague Norman Heatley improvised an ingenious apparatus to extract the compound. On 25 May 1940 eight mice received a lethal dose of bacteria; only the four also injected with penicillin lived. The Lancet, a prestigious journal, printed the outcome that August.

In February 1941 Albert Alexander, an Oxford constable gravely ill with infection, rallied on penicillin. When the scarce supply ran out, he died in March: the obstacle was quantity. From July 1941 a government laboratory in Peoria, Illinois, boosted yields about tenfold with corn-steep liquor, a starch by-product; later a mouldy market melon furnished the laboratory with a more prolific strain. By D-Day in June 1944 industrial fermentation met Allied needs.

In October 1945 Fleming, Florey, and Chain shared the Nobel prize; Heatley was overlooked. Fortune put the mould in Fleming's dish; tenacious teams and years of toil transformed it into a remedy.

中文译文

1928年9月，伦敦圣玛丽医院的细菌学家亚历山大·弗莱明度假回来，发现一团偶然出现的霉菌污染了一只培养葡萄球菌的培养皿，霉菌周围没有细菌生长。流行传说认为孢子从敞开的窗户飘入；它们其实很可能来自楼下的实验室，那里有位同事在培养霉菌。

弗莱明将霉菌产生的液汁称为青霉素，并于1929年6月发表研究。然而，粗制液汁不稳定，极难提纯；他和请来的化学家又缺乏精炼它的专业知识。他主要用它分离其他微生物，青霉素因此沉寂下来，仍是一件引人好奇的科学奇物，而非治病良药。

约十年后，在牛津，澳大利亚病理学家霍华德·弗洛里与德国出生的化学家、犹太难民恩斯特·钱恩重启研究。同事诺曼·希特利就地取材，制作了一套巧妙装置来提取这种化合物。1940年5月25日，八只小鼠接受致死剂量的细菌，只有同时注射青霉素的四只存活。权威期刊《柳叶刀》于同年8月发表了结果。

1941年2月，牛津警员艾伯特·亚历山大因感染病情危重，用上青霉素后有所好转。但稀缺的药物耗尽后，他于3月去世：难关在于产量。从1941年7月起，伊利诺伊州皮奥里亚的一家政府实验室利用淀粉生产副产物玉米浆，使产量达到原先的约十倍。后来，市场上的一只发霉甜瓜又提供了更高产的菌株。到1944年6月诺曼底登陆时，工业发酵已能满足盟军需求。

1945年10月，弗莱明、弗洛里和钱恩共同获得诺贝尔奖，希特利却被忽略了。机缘将霉菌送入弗莱明的培养皿，而多支团队锲而不舍、辛劳多年，才把它变成救人的良药。

核心词汇

先记这22项。连着下方的原文短语一起回想。

contaminate v.
污染；使混入杂质
a stray mould had contaminated

spore n.
孢子
the spores probably originated

penicillin n.
青霉素
the mould's juice penicillin

crude adj.
粗制的；未经精炼的
the crude juice

unstable adj.
不稳定的
the crude juice was unstable

purify v.
提纯；纯化
stubbornly hard to purify

expertise n.
专业知识；专长
lacked the expertise to refine it

isolate v.
分离；隔离
to isolate other microbes

revive v.
重启；使恢复活力
revived the work

improvise v.
就地取材制作；临时拼装
improvised an ingenious apparatus

ingenious adj.
巧妙的；独创的
an ingenious apparatus

apparatus n.
装置；设备
an ingenious apparatus

extract v.
提取
to extract the compound

lethal adj.
致死的
a lethal dose of bacteria

scarce adj.
稀缺的；不足的
the scarce supply

obstacle n.
障碍；难关
the obstacle was quantity

quantity n.
数量；产量
the obstacle was quantity

tenfold adv.
增至十倍
boosted yields about tenfold

prolific adj.
高产的；多产的
a more prolific strain

fermentation n.
发酵
industrial fermentation

tenacious adj.
锲而不舍的；顽强的
tenacious teams

remedy n.
良药；治疗手段
transformed it into a remedy

文中词汇与搭配

补充词汇共39项。保留每个表达在本文中的意思。

stray adj.
偶然出现的；零散的

mould n.
霉菌

bacterium n.
细菌（复数）

popular mythology 名词短语
流行传说

originate in 动词短语
起源于；来自

cultivate v.
培养；培育

stubbornly adv.
顽固地；格外难以

refine v.
精炼；提纯

chiefly adv.
主要地

microbe n.
微生物

lie dormant 动词短语
沉寂；暂未得到发展

curiosity n.
引人好奇的事物

cure n.
疗法；有效的治疗手段

chemist n.
化学家

refugee n.
难民

compound n.
化合物

dose n.
剂量

inject with 动词短语
注射（药物）

prestigious adj.
享有盛誉的

outcome n.
结果

文中词汇与搭配 续

补充词汇共39项。保留每个表达在本文中的意思。

constable n.
警员

gravely ill 形容词短语
病情危重的

infection n.
感染

rally v.
病情好转；恢复

run out 动词短语
用尽；耗尽

boost v.
提高；增加

corn-steep liquor 名词短语
玉米浆

starch n.
淀粉

mouldy adj.
发霉的

melon n.
甜瓜

furnish v.
提供；供应

with prep.
表示提供的东西

industrial adj.
工业的

Allied adj.
盟军的

overlook v.
忽视；未给予应有认可

fortune n.
机缘；运气

toil n.
辛劳；苦干

transform v.
使转化；改变

into prep.
变成；成为

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Yet the crude juice was unstable and stubbornly hard to _____, and he and the chemists he enlisted lacked the expertise to refine it.

02 Their colleague Norman Heatley _____ an ingenious apparatus to extract the compound.

03 When the scarce supply ran out, he died in March: the obstacle was _____.

04 By D-Day in June 1944 industrial _____ met Allied needs.

文章主线

- 偶然 1928年，弗莱明发现霉菌抑制细菌，把霉菌产物称作青霉素；孢子很可能来自楼下实验室，开窗说只是传闻。
- 难关 粗制青霉素不稳定、难提纯，最初多用于分离微生物；约十年后，弗洛里、钱恩和希特利等接力研究。
- 验证 8只感染小鼠中，只有用药的4只存活；1941年一名警员用药好转，却因药物耗尽死亡：瓶颈是产量。
- 量产 玉米浆使产量提高约十倍，发霉甜瓜又提供高产菌株；到1944年，工业发酵已能满足盟军用药。
- 归功 1945年弗莱明、弗洛里和钱恩获诺奖，希特利未入列；偶然发现经过多支团队多年攻关，才变成良药。

参考答案　1 purify 2 improvised 3 quantity 4 fermentation